Zhiying Cui

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EDUCATION

Ningbo University | B.Eng. in Computer Science and Technology — **Aug. 2022 – Jun. 2026**
GPA: 3.54/4.00   Average Score: 87/100 — *Ningbo, China*

PUBLICATIONS

Conference Papers

C1 **Z. Cui**, L. Gao, M. Yang, Y. Dong, and P. Li. "SciFigPlag-Bench: A Benchmark for Provenance-Aware Scientific Figure Plagiarism Detection," *Conference on Empirical Methods in Natural Language Processing (EMNLP 2026), Submitted.* [PDF]

C2 M. Yang, P. Li*, L. Gao*, **Z. Cui**, W. Li, and J. Xiao. "Decay Pruning Method: Smooth Pruning With a Self-Rectifying Procedure," *IEEE International Conference on Computer Vision (ICCV) Workshops*, 2025, pp. 3117-3123. [PDF]

Journal

J1 **Z. Cui**, L. Gao, M. Yang, D. Wang and Y. Dong. (2024). "Global–Local Feature Fusion for Glass Curtain Wall Damage Detection ," *IEEE Transactions on Geoscience and Remote Sensing, Unsubmitted manuscript, 2026.*

Patents

P1 **Z. Cui**, M. Yang, D. Wang, H. Cao, and L. Gao. (2024). "Autonomous Detection Method for High-altitude Glass Curtain Wall Damage Areas Based on Unmanned Aerial Vehicles," China National Intellectual Property Administration, Patent No. ZL 2024 1 0449517.2.

P2 C. Lin, L. Gao, Y. Shou, J. Chen, and **Z. Cui**. (2024). "A Multi-class Classification Method for Multi-center sMRI Images Based on Generative Adversarial Networks," China National Intellectual Property Administration, Patent No. ZL 2023 1 1261313.8.

P3 S. Zhu, L. Gao, S. Lan, C. Lin, Y. Shou, and **Z. Cui**. (2024). "Multi-center MRI Image Classification Method," China National Intellectual Property Administration, Patent No. ZL 2023 1 1261338.8.

ACADEMIC SERVICE

CVPR 2026 Workshop Reviewer — **2026**
Workshop on Emerging Directions in Data for Multimodal Foundation Models

RESEARCH EXPERIENCE

SciFigPlag-Bench: A Benchmark for Provenance-Aware Scientific Figure Plagiarism Detection, — **Apr. 2026 – May 2026**
Co-advised by Pengyuan Li(MIT-IBM Watson AI Lab Researcher) and Prof. Linlin Gao — *Ningbo University, China*

- Built the **SciFig-Plag** dataset, including real-world plagiarized figures, synthetically generated samples, and negative examples retrieved via web search.
- Defined a taxonomy of figure reuse types, designed evaluation tasks, and implemented annotation and benchmarking pipelines for scientific figure plagiarism detection.

Global–Local Feature Fusion for Glass Curtain Wall Damage Detection — **Dec. 2024 – Present**
Advised by Prof. Linlin Gao — *Ningbo University, China*

- Proposed a global–local fusion framework: used Grad-CAM to locate attention regions, cropped high-resolution local patches, extracted fine-grained detail features, and fused them with global context for accurate damage detection.
- Achieved 95.4% precision, outperforming YOLOv11, YOLOv12, and YOLOv26 baselines.

Efficient Path Planning Algorithm Based on Bidirectional A* — **Sep. 2024 – Dec. 2024**
Advised by Prof. Linlin Gao — *Ningbo University, China*

- Proposed an enhanced Bidirectional A* search algorithm using the Manhattan-distance heuristic, achieving a 14-fold speedup (from 0.71s to 0.05s) with path errors under 2 pixels.

- Integrated a dynamic color-thresholding method for real-time robustness and led a 3-member team to first place among all undergraduate and graduate teams, with a score of 94/100.

Decay Pruning Method: Pruning Smoothly with a Self-Rectifying Procedure **Jun. 2023 – Jun. 2024**
Advised by Prof. Linlin Gao *Ningbo University, China*

- Proposed the **Decay Pruning Method (DPM)**, which gradually decays weights during training and uses gradient feedback for smooth, self-rectifying pruning.
- Improved post-pruning performance and stability by mitigating accuracy loss during dynamic network evolution.

Multi-Center MRI Image Classification Based on CNNs **Sep. 2023 – May 2024**
Advised by Prof. Linlin Gao *Ningbo University, China*

- Proposed a CNN-based model to address inter-center data heterogeneity and the limitations of manual feature engineering.
- Improved model generalization and robustness across multi-center clinical datasets through automatic feature learning and a specialized distribution-alignment strategy.

INDUSTRY EXPERIENCE

Loctek Ergonomic Technology Corp **Jul. 2025 – Sep. 2025**
Algorithm Intern, mentored by Chengbin Peng *Ningbo, China*

- Refactored core C++ modules (for calibration and coordinate transformation) into Python, resulting in significant improvements in module maintainability and scalability.
- Achieved precise stereo 3D point-cloud alignment by resolving post-transformation misalignment using spatial mathematics.
- Utilized optical flow methods for sequential image stabilization, successfully ensuring accurate target centering.

COMPETITIONS

2024 COMAP Mathematical Contest in Modeling | Finalist (Top 1% ∼ 2%)

2025 COMAP Mathematical Contest in Modeling | Meritorious (Top 7% ∼ 8%)

2025 “The Challenge Cup” | National Third Prize

FUNDED PROJECTS

2025 “Zhejiang New Talent Program” Project
2024 “Zhejiang New Talent Program” Project
2023 Student Research and Innovation Program (SRIP) Research Project

AWARDS

2026 Loctek Dean’s Scholarship
2025 Provincial Government Scholarship
2025 Loctek Innovation and Entrepreneurship Scholarship, Third Prize
2025 Jiang Xinghao Student Scientific Research Achievement Award, First Prize
2024 Provincial Government Scholarship
2023 University Special Scholarship

SKILLS

Programming Languages : Python, C/C++, JavaScript, SQL, MATLAB
Frameworks & Tools : PyTorch, Vue.js, OpenCV, LaTeX, Linux, GitHub, VS
Code **Languages** : Chinese(Native), Korean(TOPIK 6), English(TOEIC 780)